Andrew Strominger and David Morrison showed that a brane can wrap itself around a spherical portion of a Calabi-Yau shape, much like plastic wrap vacuum-sealed around a grapefruit; should that portion of space shrink, the wrapped brane would also shrink, causing its mass to decrease. This decrease in mass, we were able to show, allows the portion of space to collapse fully and tear open—space itself can rip apart—while the wrapped brane ensures that there are no catastrophic physical consequences. I discussed this development in detail in *The Elegant Universe* and will briefly return to it when we discuss time travel in Chapter 15, so I won't elaborate further here. But this snippet makes clear how higher-dimensional branes can have a significant effect on the physics of string theory.

For our current focus, though, there is another profound way that branes impact the view of the universe according to string/M-theory. The grand expanse of the cosmos—the entirety of the spacetime of which we are aware—may itself be nothing but an enormous brane. Ours may be a braneworld.

Braneworlds

Testing string theory is a challenge because strings are ultrasmall. But remember the physics that determined the string's size. The messenger particle of gravity—the graviton—is among the lowest-energy string vibrational patterns, and the strength of the gravitational force it communicates is proportional to the length of the string. Since gravity is such a weak force, the string's length must be tiny; calculations show that it must be within a factor of a hundred or so of the Planck length for the string's graviton vibrational pattern to communicate a gravitational force with the observed strength.

Given this explanation, we see that a highly energetic string is not constrained to be tiny, since it no longer has any direct connection to the graviton particle (the graviton is a *low*-energy, zero-mass vibrational pattern). In fact, as more and more energy is pumped into a string, at first it will vibrate more and more frantically. But after a certain point, additional energy will have a different effect: it will cause the string's length to increase, and there's no limit to how long it can grow. By